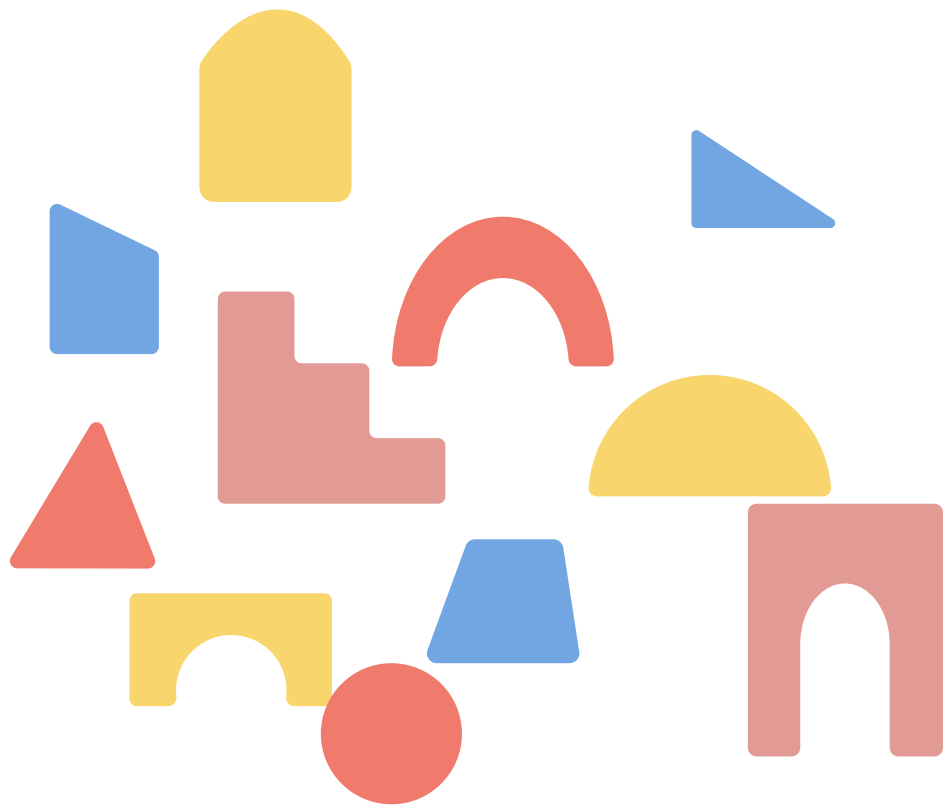




DADA HUB RESIDENCY • APRIL 23, 2026

BUILDING USEFUL AI ASSISTANTS FAST

FROM API ACCESS TO A SIMPLE RAG CHATBOT

SPEAKER

CEDROUSEROLL OMONDI

13:00-14:00 EAT

FOCUS

PRACTICAL AI BUILDING

API KEYS, PROMPTS, CHATBOT, RETRIEVAL, COSTS



TODAY

WHAT YOU'LL LEAVE WITH



API access

OpenAI / Claude setup basics

Prompt structure

Role, task, context,
constraints

Chatbot flow

UI, API route, model
response

Embeddings

Search by meaning

Simple RAG

Retrieve, ground, answer

Cost thinking

Build within real constraints



Working starting point, not theory overload.



USE CASE

ONE EXAMPLE, END TO END

DADA DEVS PROGRAM ASSISTANT

RELATABLE

Cohort applications, program pathways, Dada Hub questions, mentor info

FLEXIBLE

Works with multiple personas and a small knowledge base

PRACTICAL

Easy to continue during the self-directed block

Prompt



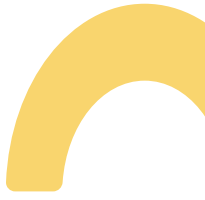
Data



Model



UI



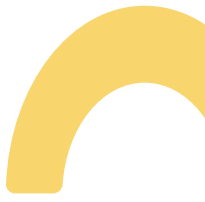


SETUP

API ACCESS IN PRACTICE

- 
- Get a provider key
 - Store it in `.env.local`
 - Keep secrets server-side
 - Test with one request

```
OPENAI_API_KEY=your_key_here  
OPENAI_MODEL_CHEAP=gpt-4.1-nano  
OPENAI_MODEL_COMPLEX=gpt-4.1-mini  
EMBEDDING_MODEL=text-embedding-3-small
```



PROMPTING

A USEFUL PROMPT HAS STRUCTURE



Role

Who the assistant is

Task

What it should do

Context

What situation it is in

Constraints

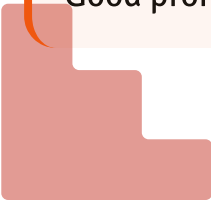
Length, honesty, limits

Output style

Tone and format

Formula

Role + Goal + Context +
Constraints + Style



Good prompt = Role + Goal + Context + Constraints + Output style



WEAK VS BETTER PROMPT

WEAK PROMPT

PROMPT

Help users with Dada Devs.

WHAT'S MISSING

- No assistant role
- No audience or use case context
- No rules about uncertainty
- No required output structure

WEAK VS BETTER PROMPT

BETTER PROMPT STRUCTURE

PROMPT

Role: You are the Dada Devs Program Guide.

Task: Help users understand pathways, eligibility, Dada Hub, workshops, mentorship, and partner opportunities.

Context: Answer using retrieved Dada Devs website content when available. If the site content does not support a claim, say so.

Constraints: Keep answers concise, practical, and accurate. Do not invent dates, requirements, or promises.

Output format: Use short sections: `Answer` , `Recommended next step` , and `Sources used` .

WEAK VS BETTER PROMPT

BETTER PROMPT EXAMPLE

EXAMPLE INPUT

How can I move from learning into Bitcoin open-source contribution?

EXAMPLE OUTPUT

Answer: Start with the pathway that matches your level, then use Dada Hub and mentorship opportunities to move into practical contribution.

Recommended next step: Review the Pathways page and the Bitcoin track.

Sources used: Pathways, Dada Hub.

PERSONAS

SAME MODEL, DIFFERENT ASSISTANT

PERSONA 1

PROGRAM GUIDE

Clear, structured, pathway-focused

PERSONA 2

RESIDENCY SUPPORT

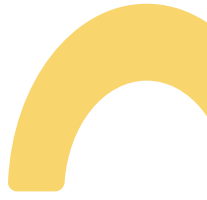
Helpful, schedule-aware, operational

PERSONA 3

CONTRIBUTOR GUIDE

Contribution-oriented, technical, next-step driven

Prompting is product design.



ARCHITECTURE

THE SIMPLEST USEFUL CHATBOT FLOW

User



Next.js UI



API route



Model



Response

KEEP

- One page
- One API route
- One model call
- One persona config

SKIP FOR NOW

- Auth
- Database
- Agents
- Complex orchestration



DEMO

WHAT THE APP IS DOING



FRONTEND

1. Select persona
2. Type message
3. Send request
4. Render answer

BACKEND

1. Load persona prompt
 2. Retrieve relevant context
 3. Call the model
 4. Return grounded response
- 
- 

BEFORE LLMS

HOW CHATBOTS WORKED BEFORE GPT-STYLE MODELS

Tokenize text



TF-IDF vectors



Intent classifier



Pre-written response

TYPICAL PIPELINE

1. Split text into words, sentences, or paragraphs
2. Vectorize with TF-IDF across the corpus
3. Predict intent with models like Naive Bayes
4. Select a predefined answer for that intent

MAIN LIMITATION

- Rigid response set
- Weak handling of new phrasing
- No real synthesis across documents
- Heavy manual maintenance

INDEXING

HOW WE BUILD THE KNOWLEDGE BASE

Scrape Dada Devs pages



Chunk sections



Create embeddings



Store site index

SOURCE PAGES

- Home, About, Pathways, Workshop, Contact
- Dada Citadel, Mentor, Partner pages
- Bitcoin track curriculum
- Blog posts and related internal pages

WHY THIS MATTERS

- The chatbot answers from current org content
- We can inspect exactly what was indexed
- We avoid one giant prompt with all site text
- Retrieval becomes teachable, not hidden



EMBEDDINGS

WHAT EMBEDDINGS ARE DOING HERE



1. CONVERT

Each text chunk becomes a numeric vector using `text-embedding-3-small`.

2. COMPARE

The user query is embedded too, then compared against all chunk vectors.

3. RETRIEVE

The closest chunks by meaning are passed to the model as grounded context.



Embeddings are not the answer. They are the search layer that helps us find the right context cheaply.



EMBEDDINGS

FROM SCRAPED TEXT TO SEARCHABLE CHUNK

1. SCRAPED TEXT

Source page: Mastering Bitcoin track

Bitcoin Core Development Environment
Configuring the Core Build
Running a Bitcoin Node
A developer is needed for this session

2. CHUNKED RECORD

Title: Mastering Bitcoin — Bitcoin Core

Section: Bitcoin Core

Stored with URL, page title, section, tags, and chunk text.

3. CONVERTED

Embedding vector:

```
[0.014, -0.083, 0.221, 0.057,  
... ]
```

The vector is long, but now this chunk can be compared by meaning instead of exact words.

This is the key move: website text becomes structured records, then vectorized search targets.

RAG

FROM CHATBOT TO GROUNDED ASSISTANT

Question



Embed query



Retrieve top-k chunks



Pack context



Generate answer

RETRIEVAL STEP

We embed the user query, score it against the site index, and keep only the highest-similarity chunks.

GROUNDING STEP

We inject retrieved chunks with page title, section, and URL so the model has bounded context.

GENERATION STEP

The LLM synthesizes an answer from the retrieved evidence instead of relying only on parametric memory.

RAG

WHAT ACTUALLY GOES INTO THE MODEL CALL

SYSTEM PROMPT

- Persona instructions
- Closed vs open scope rules
- Use retrieved Dada Devs context when relevant
- Say when support is missing

RETRIEVED CONTEXT BLOCK

- Source title
- Page and section labels
- URL
- Chunk text



EMBEDDINGS

SEMANTIC SEARCH



USER ASKS

**HOW CAN I MOVE FROM LEARNING
INTO REAL BITCOIN OPEN-SOURCE
CONTRIBUTION?**

SYSTEM RETRIEVES

**PATHWAYS, DADA HUB, MENTOR
AND PARTNER PAGES**



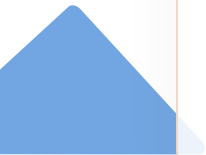
The retrieved pages do not need to repeat the same exact words. They just need to be close in meaning.





EMBEDDINGS

KEYWORD SEARCH VS SEMANTIC SEARCH



WHY KEYWORD SEARCH IS WEAK

The best source might not literally contain the same words as the user query.

WHY SEMANTIC SEARCH HELPS

The app can connect related ideas like pathways, contribution, residency, curriculum, and mentorship.

Keyword search

Finds exact terms

Semantic search

Finds related meaning

Embedding match

Shows which chunks are likely worth paying to send





COSTS

BUILD FOR REAL CONSTRAINTS

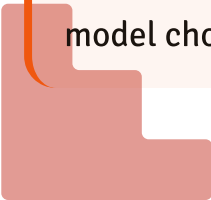


WHAT INCREASES COST

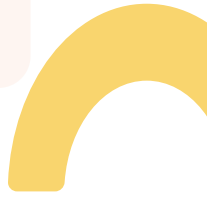
- Too much context
- Larger models by default
- Long responses
- Too many retrieved chunks

WHAT REDUCES COST

- Use smaller models first
- Limit tokens
- Retrieve only the top 2–4 relevant chunks
- Cache repeated answers



Optimize for token budgets, mobile realities, and bandwidth limits. In the demo, retrieval depth and model choice both change with query complexity.





NEXT

SELF-DIRECTED BLOCK + HOMEWORK

BUILD A CHATBOT FOR A LOCAL USE CASE



EXAMPLES

- Dada Devs cohort application assistant
- Dada Hub residency schedule assistant
- Bitcoin track FAQ assistant
- Mentor and partner inquiry assistant
- Open-source contribution guide

KEEP SCOPE SMALL

- One audience
 - One persona
 - One knowledge base
 - One simple UI
- 



RESOURCES

TOOLS AND REFERENCES USED



Next.js

App Router full-stack workshop app

nextjs.org/docs/app/getting-started/installation

shadcn/ui

UI primitives and registry workflow

ui.shadcn.com

AI SDK

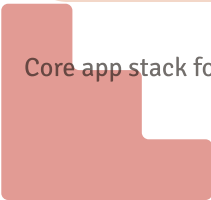
Chat route, streaming, UI integration

ai-sdk.dev/docs

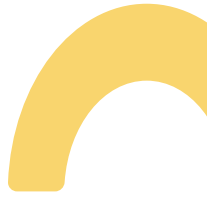
assistant-ui

Chat interface primitives and timing UI

assistant-ui.com/examples/ai-sdk



Core app stack for the practical: framework, UI layer, streaming, and chat interface.





RESOURCES

WORKFLOW AND DELIVERY REFERENCES



OpenAI API

Model calls and embeddings

platform.openai.com/docs

Slidev

Presentation authoring and delivery

sli.dev

MCP

Mentioned as part of the modern AI tooling ecosystem

modelcontextprotocol.io

Codex + Skills

Coding agent workflow using Slidev, Next.js, shadcn, and frontend design skills

OpenAI Codex coding agent



Also used in this workflow: semantic retrieval, persona prompting, cost-aware routing, and local knowledge-base chunking.



RESOURCES

EMBEDDINGS AND OPEN MODELS

OpenAI Embeddings Guide

Official guide for creating and using embeddings

platform.openai.com/docs/guides/embeddings

Sentence Transformers

Strong open-source starting point for semantic search from scratch

sbert.net/docs/quickstart.html

Hugging Face Hub

Explore pretrained models, embedding models, datasets, and transformers

huggingface.co/docs/hub/models

Ollama

Run open models locally, including Llama-family and other small models

docs.ollama.com



BUILD SOMETHING USEFUL.

KEEP IT SMALL.

**MAKE IT CONTINUE AFTER
TODAY.**

